

Report 11/13/2020

P. Hacker

HELMHOLTZ
SPITZENFORSCHUNG FÜR
GROSSE HERAUSFORDERUNGEN



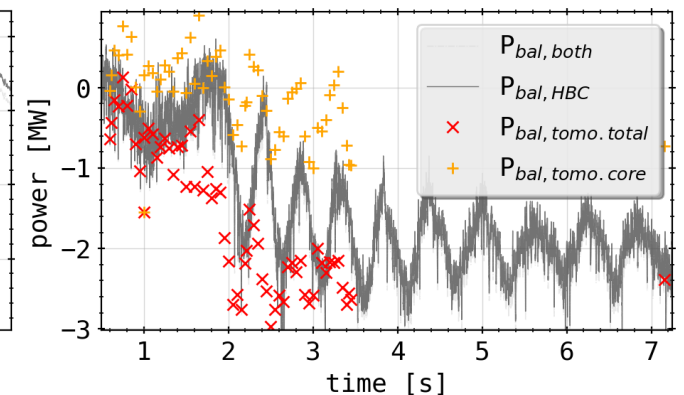
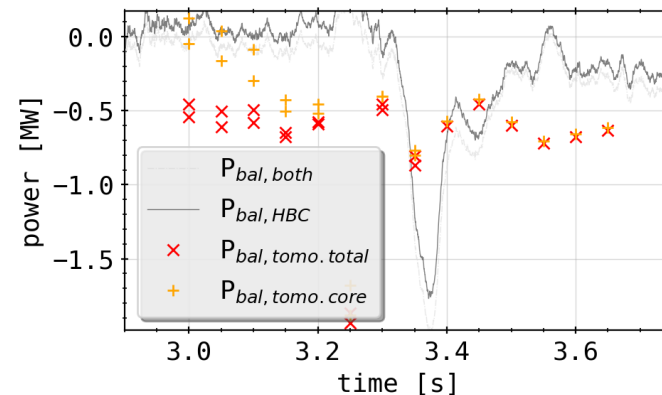
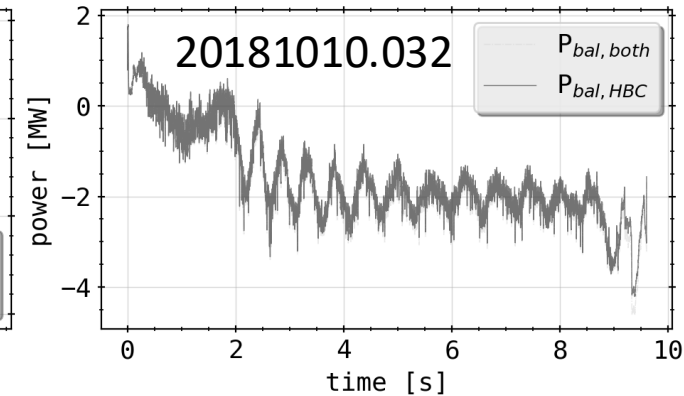
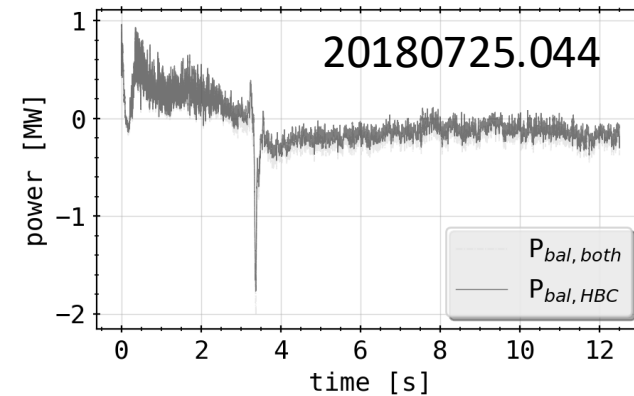
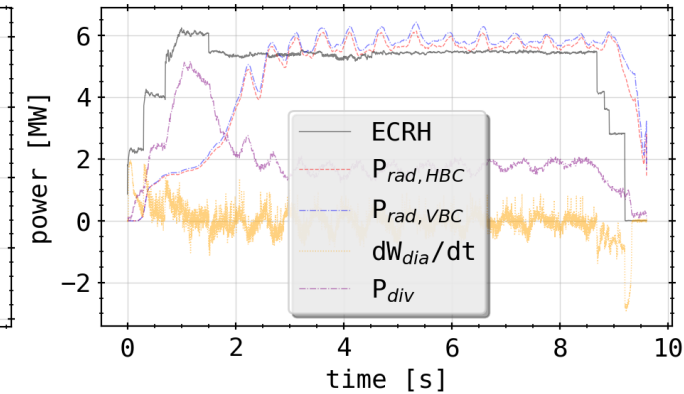
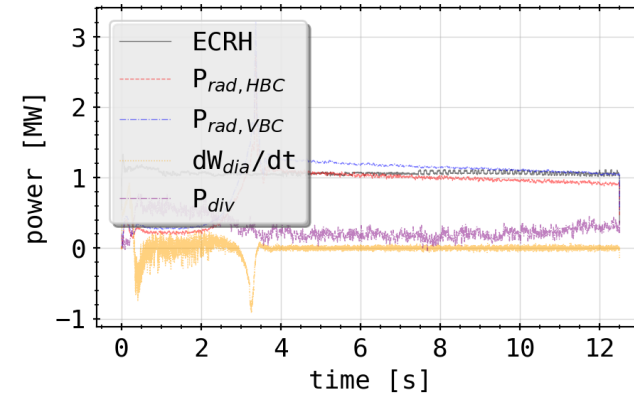
This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 and 2019-2020 under grant agreement No 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Tomogram Statistics: Experiment Power Balance

- compiled experiment data and parameters to calculate

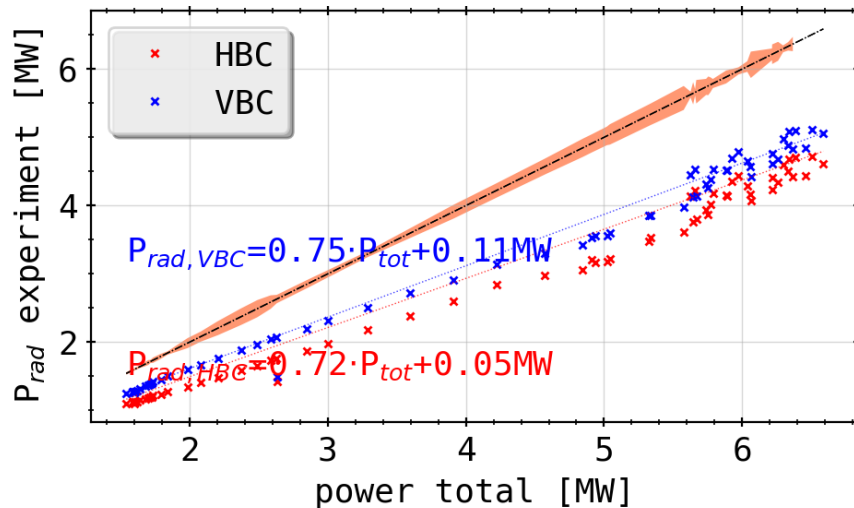
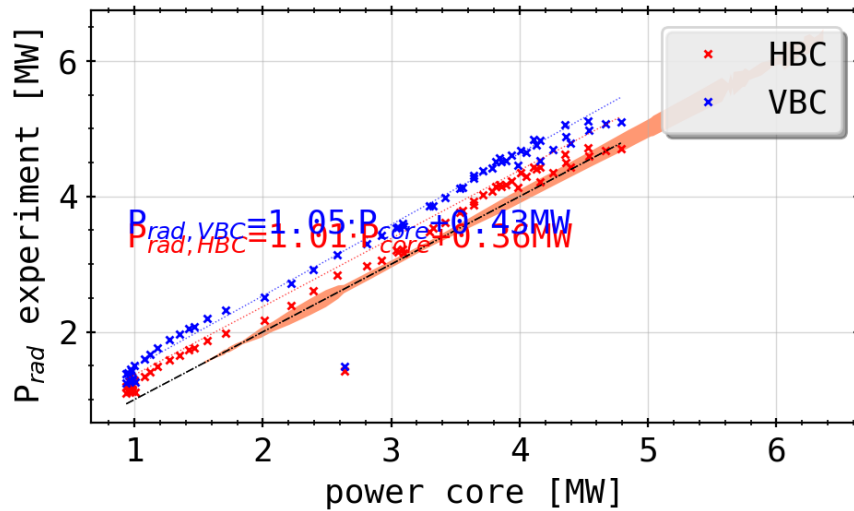
$$P_{bal}(t) = P_{ECRH}(t) - P_{rad,HBC/VBC}(t) - P_{div}(t) - \frac{dW_{dia}}{dt}(t)$$

- also exchanged radiated power proxy from bolometer camera by 2D integrated power of tomogram (both core and total)



Tomogram Statistics: Experiment Reconstructions Errorbar

20181010.032

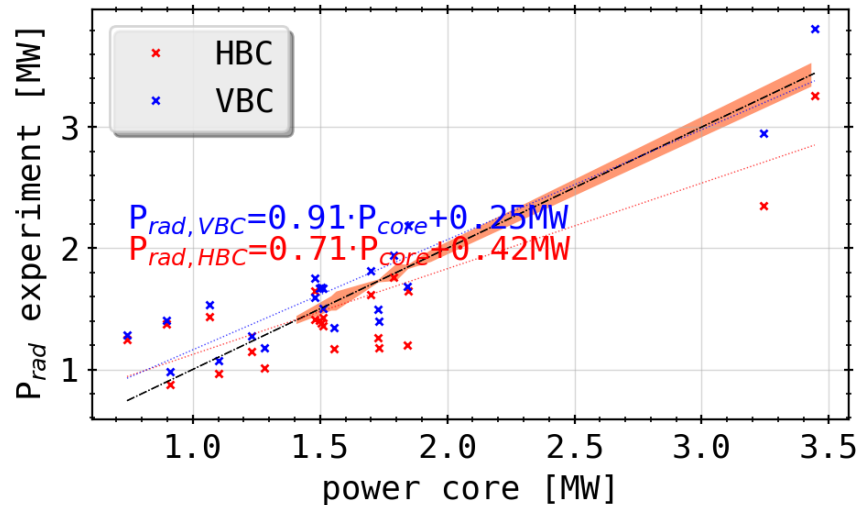


- extra slope in P_{rad} vs. total power by cutting negative results from reconstruction when limiting to only physically meaningful numbers ($P/m^3 > 0$)?
- increase of their impact is expected with overall higher radiation level to counteract errors in chordal profile

CONCLUSION

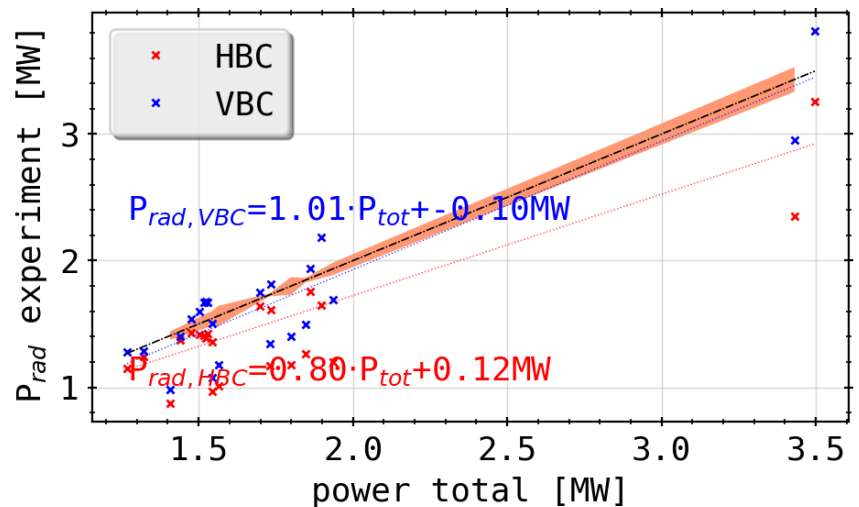
- impact of mathematical error is negligible across the power spectrum, but why?
- fit of integrated power values vs camera proxy yields interesting results ($\sim 0.4 \text{ MW}$ offset to core power, ~ 0.75 ratio to total 2D power)

20180725.044

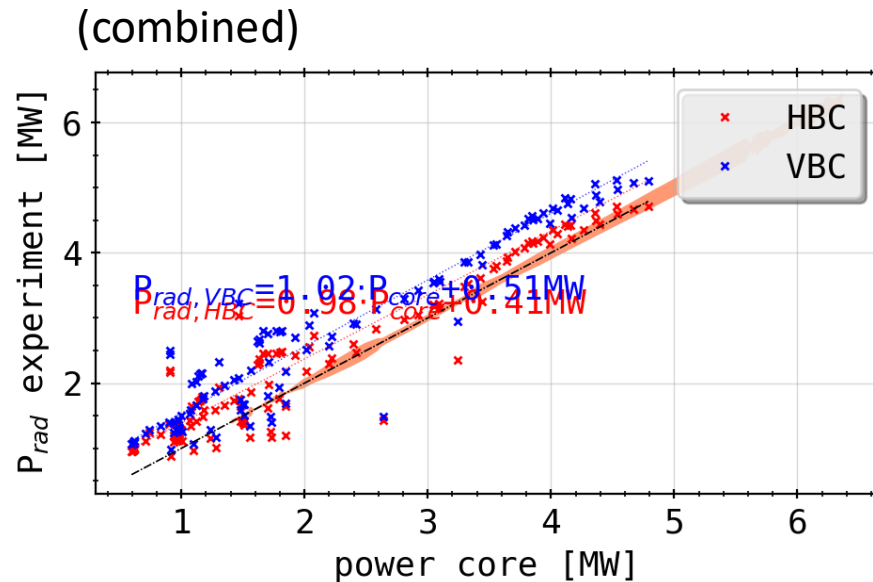


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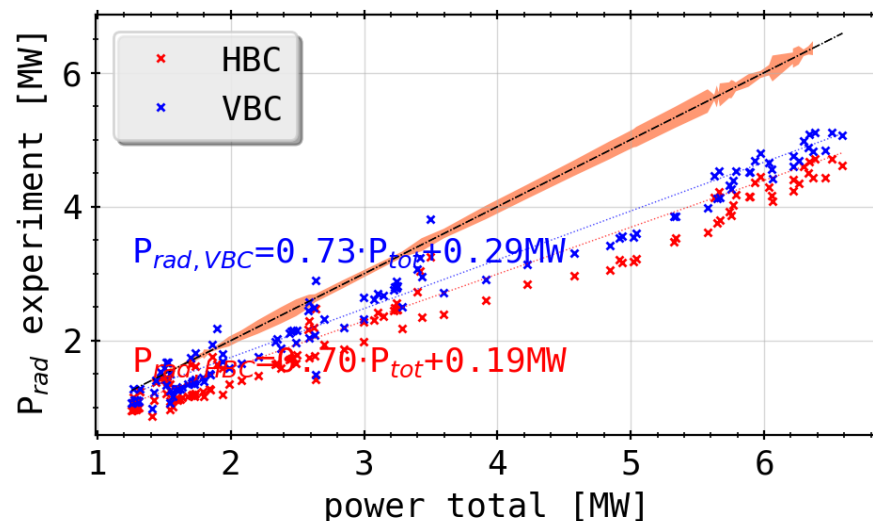


- impact of mathematical error is negligible across the power spectrum, but why?
- fit of integrated power values vs camera proxy yields interesting results ($\sim 0.4 \text{ MW}$ offset to core power, ~ 0.7 ratio to total 2D power)
- changes for different discharge types?



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- increase of their impact is expected with overall higher radiation level to counteract errors in chordal profile

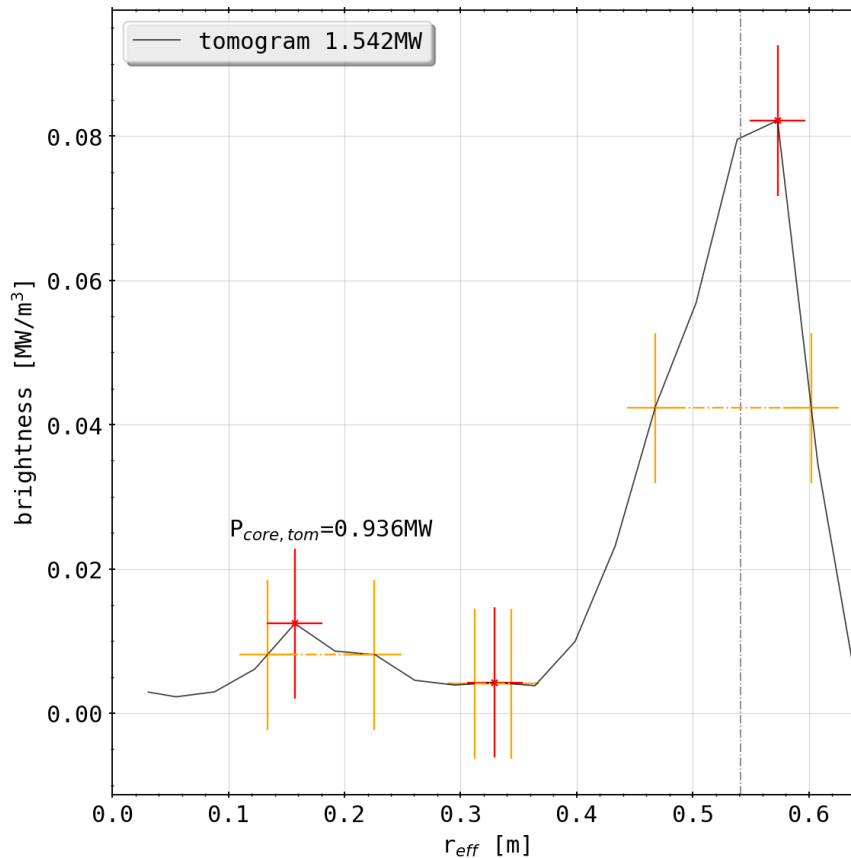
CONCLUSION



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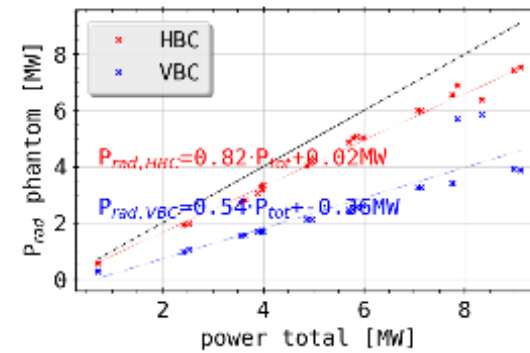
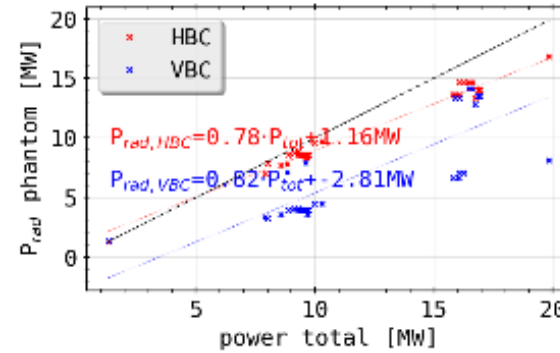
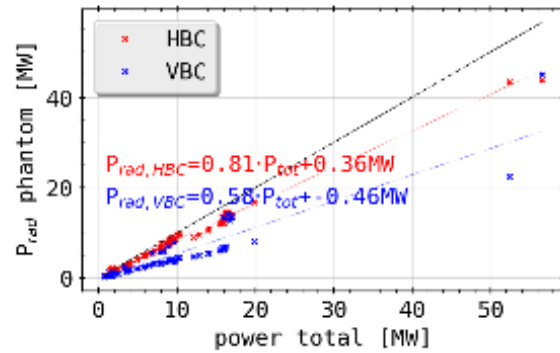
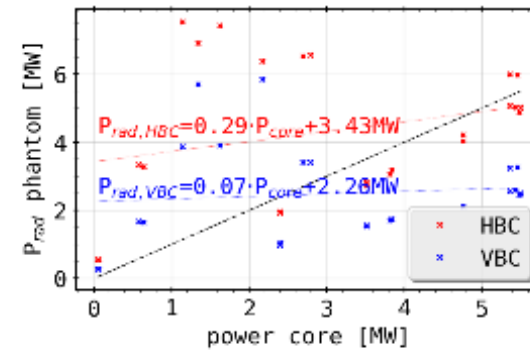
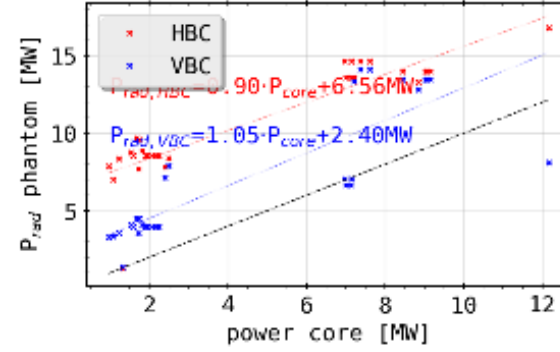
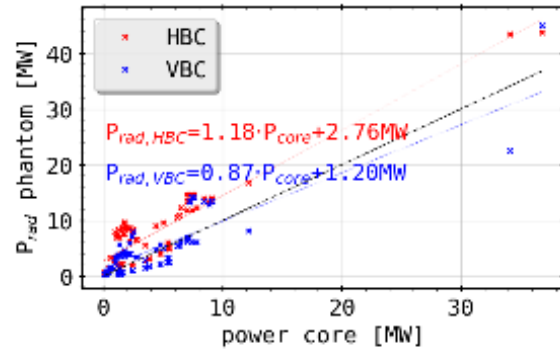
Tomogram Statistics: Experiment Reconstructions Errorbar

20180725.044@3.45s



- errorbar looks large here, because the total error for the tomogram (minimum value < 0.0 can be single cell or whole 'ring') has to assumed possible at every integration point
- e.g.: at powers (both $X^2 \sim 1.0$)
 - total: 5.78 MW , neg: 66.63 kW
 - total: 6.33 MW, neg: 61.68 kW
 - total: 2.37 MW, neg: 121.51 kW

Tomogram Statistics: Phantom Reconstructions

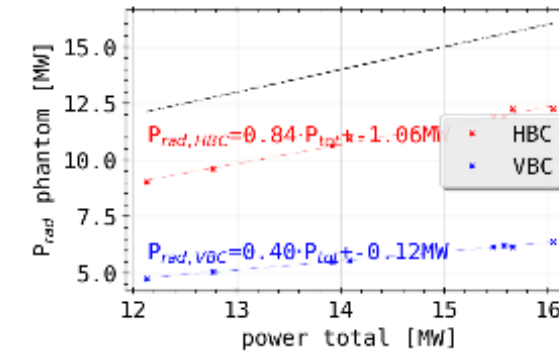
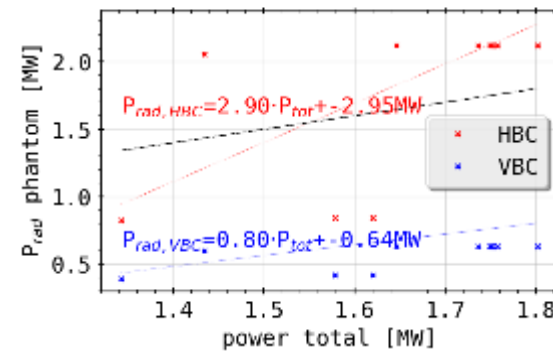
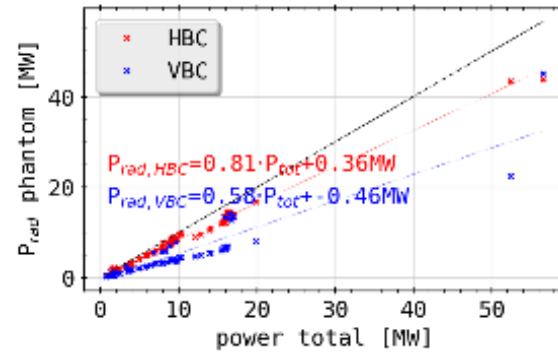
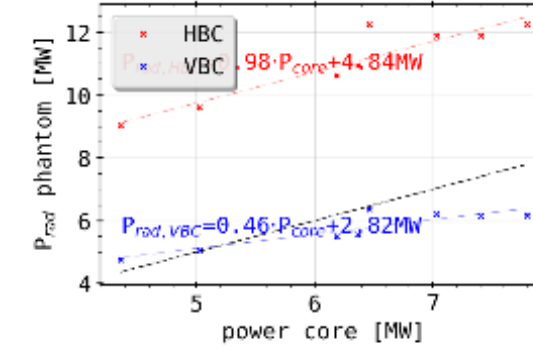
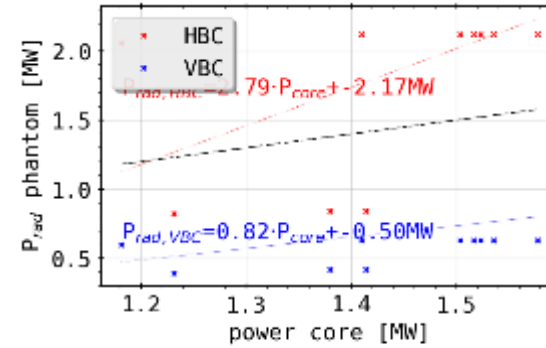
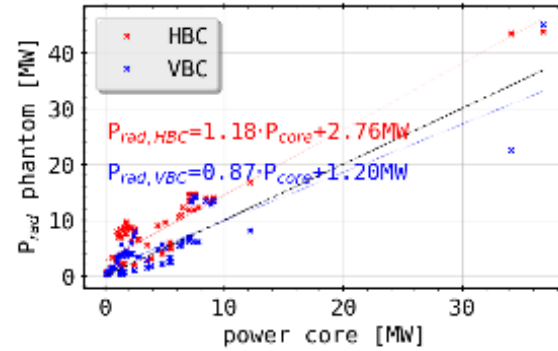


(all phantoms)

(island mimics)

(radiation ring)

Tomogram Statistics: Phantom Reconstructions



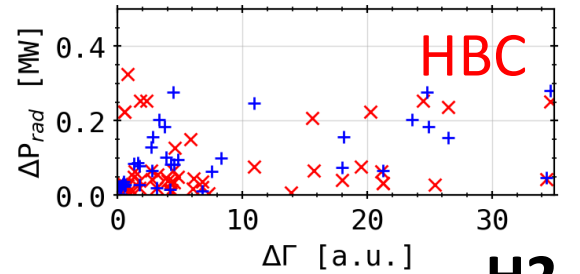
(all phantoms)

(radiation spot)

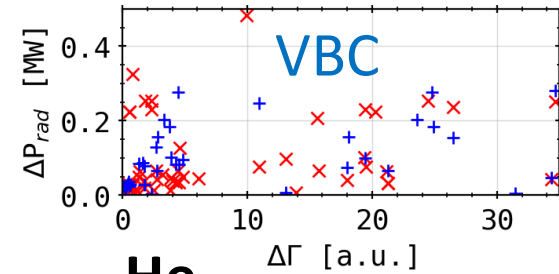
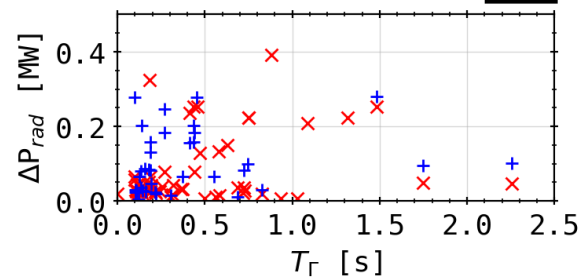
(double ring)

Peak Statistic: Methane Session with QSQ

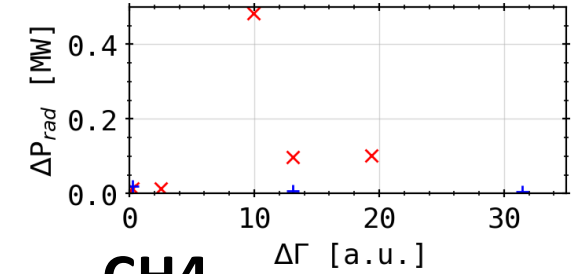
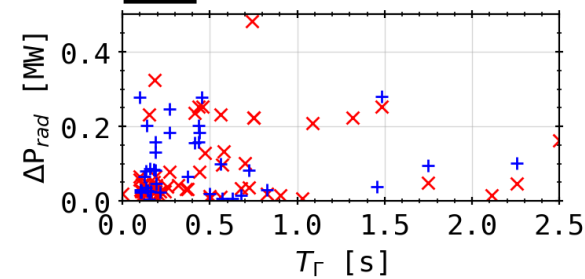
100 ms



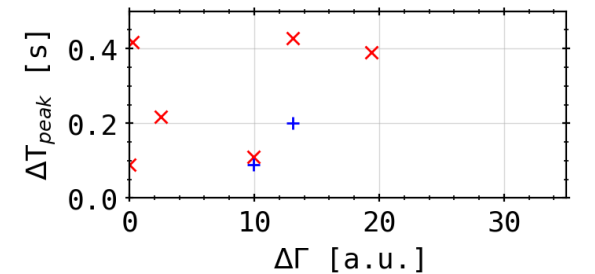
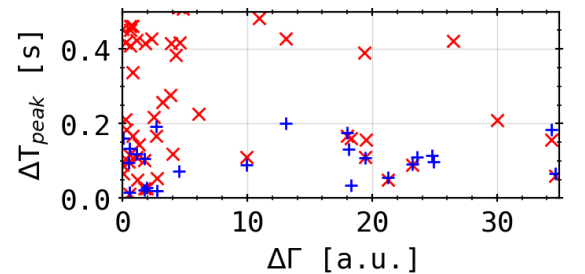
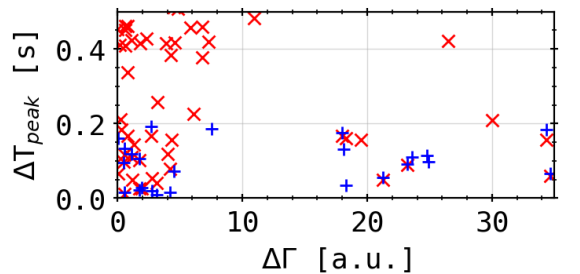
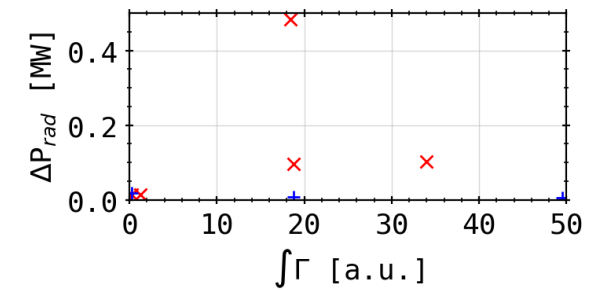
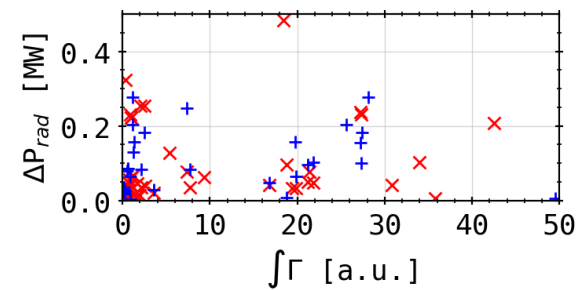
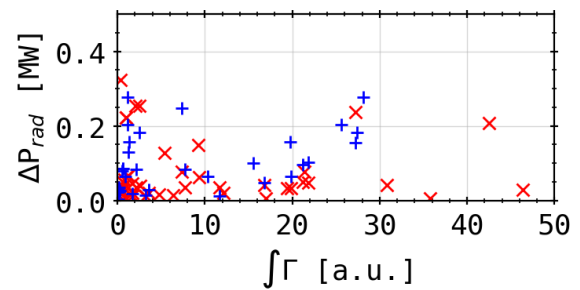
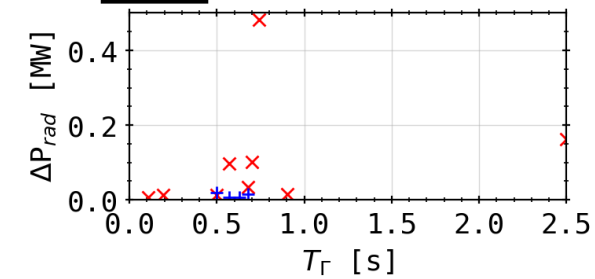
H2



He

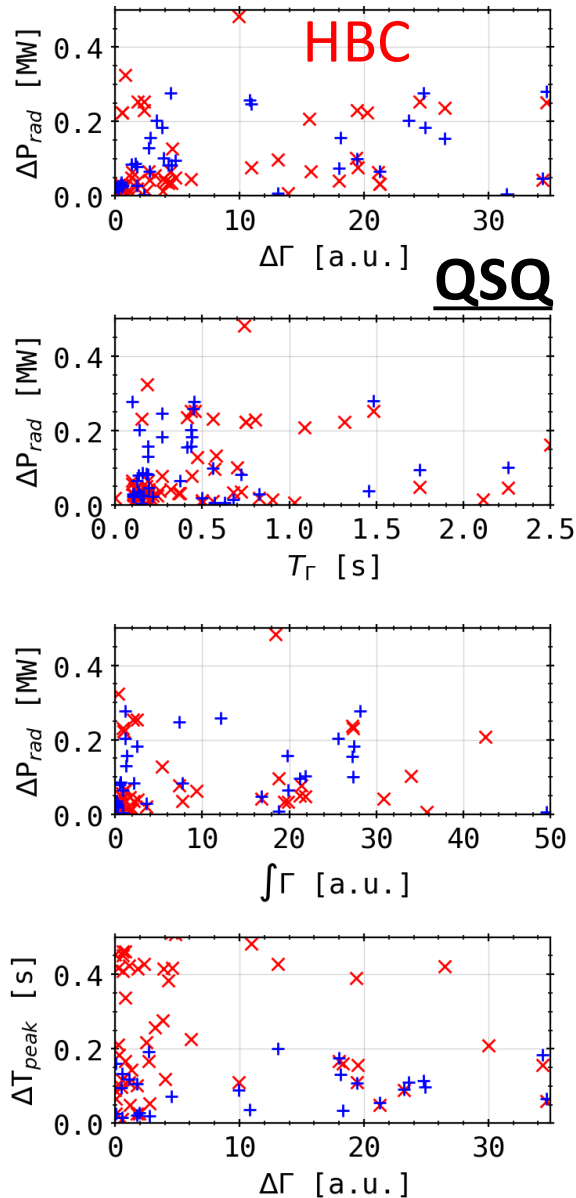


CH4



Peak Statistic: Methane Session with QSQ

100 ms



Main
Valve

